DA-23 Salary Band & Pay Gap Analysis

Step 3: SQL Analysis (HR & Workforce)

Objective

Answer the business question **"Is pay consistent across roles and gender?"** using SQL on the cleaned employee dataset (10,500 employees, 12 cities, 8 departments, 8 job roles). Salaries are annual CTC in Indian rupees (1 lakh = 100,000; 1 crore = 10,000,000).

Database design (4 related tables)

Table	Key columns	Rows
employees (fact)	emp_id (PK), dept_id, role_id, city_id (FKs), gender, age, education, experience_years, joining_date, annual_salary, performance_rating, salary_imputed, salary_band	10,500
departments	dept_id (PK), dept_name	9
job_roles	role_id (PK), role_name, job_level	8
cities	city_id (PK), city_name	13

The cleaned CSV was split into these tables so that queries use real JOINs. Relationship: employees.dept_id → departments, employees.role_id → job_roles, employees.city_id → cities.

SQL techniques covered

Technique	Used in
JOIN (2, 3 and 4 tables)	Q1, Q2, Q4, Q5, Q6, Q7, Q8
GROUP BY / HAVING / CASE	Q1, Q2, Q3, Q5, Q6, Q10
Subquery	Q3, Q6, Q7
CTE (WITH)	Q4, Q9
Window functions (ROW_NUMBER, RANK, NTILE, AVG OVER)	Q7, Q8, Q9

Note: Queries were run and verified on the full data. Syntax is standard SQL that runs on MySQL 8+ and PostgreSQL. Salaries for 536 employees were imputed during cleaning (flagged in salary_imputed); results are not materially affected.

Q1. Headcount and average salary by department

Business question: Which departments pay the most and how big are they? [JOIN + GROUP BY]

```
SELECT d.dept_name, COUNT(*) AS headcount,
       ROUND(AVG(e.annual_salary)) AS avg_salary,
       MIN(e.annual_salary) AS min_salary, MAX(e.annual_salary) AS max_salary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY avg_salary DESC;
```

Result

dept_name	headcount	avg_salary	min_salary	max_salary
Data Science	855	1,885,490	339,900	11,373,800
IT	2295	1,662,593	278,500	9,732,600
Finance	1184	1,555,915	264,000	8,109,000
Unknown	98	1,427,314	268,100	9,787,400
Marketing	985	1,272,855	213,000	7,395,400
Sales	1825	1,266,235	201,400	7,243,400
HR	844	1,149,334	221,300	6,276,400
Operations	1554	1,131,181	193,600	5,950,600
Customer Support	860	865,825	156,700	4,339,000

Insight: Data Science pays the most (avg Rs 18.9 lakh) and Customer Support the least (Rs 8.7 lakh), a 2.2x spread. IT, Sales and Operations hold the largest headcount. 98 employees have an Unknown department and should be fixed at source.

Q2. Salary band (min / average / max) for each job role

Business question: What is the salary band for every role? Is the band wide or tight? [JOIN + GROUP BY]

```
SELECT r.job_level, r.role_name, COUNT(*) AS employees,
       MIN(e.annual_salary) AS band_min,
       ROUND(AVG(e.annual_salary)) AS band_avg,
       MAX(e.annual_salary) AS band_max,
       ROUND(MAX(e.annual_salary) * 1.0 / MIN(e.annual_salary), 2) AS max_to_min_ratio
FROM employees e
JOIN job_roles r ON e.role_id = r.role_id
GROUP BY r.job_level, r.role_name
ORDER BY r.job_level;
```

Result

job_level	role_name	employees	band_min	band_avg	band_max	max_to_min_ratio
1	Intern	432	156,700	303,241	542,500	3.46
2	Junior Associate	2361	277,500	567,686	1,061,200	3.82
3	Associate	2681	369,900	858,020	1,589,200	4.3
4	Senior Associate	2142	531,100	1,242,448	2,454,500	4.62
5	Team Lead	1235	791,200	1,772,696	3,415,000	4.32
6	Manager	786	1,119,900	2,467,497	4,996,100	4.46
7	Senior Manager	550	1,759,600	3,776,577	7,187,000	4.08
8	Director	313	2,895,000	5,776,799	11,373,800	3.93

Insight: Bands are very wide: the highest paid person in a role earns 3.5x to 4.6x the lowest. Adjacent bands overlap heavily (an Associate can earn Rs 15.9 lakh while a Senior Associate can earn Rs 5.3 lakh), so salary bands are not being enforced.

Q3. Overall gender pay gap

Business question: How much less do women earn than men on average? [GROUP BY + subquery]

```
SELECT gender, COUNT(*) AS employees,
       ROUND(AVG(annual_salary)) AS avg_salary,
       ROUND((1 - AVG(annual_salary) /
              (SELECT AVG(annual_salary) FROM employees WHERE gender = 'Male')) * 100, 2) AS gap_vs_male_pct
FROM employees
WHERE gender IN ('Male', 'Female')
GROUP BY gender;
```

Result

gender	employees	avg_salary	gap_vs_male_pct
Female	3943	1,326,395	6.24
Male	6236	1,414,717	0

Insight: Women earn 6.24% less than men on average (Rs 13.26 lakh vs Rs 14.15 lakh), about Rs 88,000 per year. This is the headline pay gap.

Q4. Gender pay gap within the SAME job role

Business question: Is the gap real, or only because women hold lower-level roles? [CTE + JOIN]

```
WITH role_gender AS (
    SELECT role_id, gender, AVG(annual_salary) AS avg_sal, COUNT(*) AS n
    FROM employees
    WHERE gender IN ('Male', 'Female')
    GROUP BY role_id, gender
)
SELECT r.role_name,
       ROUND(m.avg_sal) AS male_avg, ROUND(f.avg_sal) AS female_avg,
       ROUND((m.avg_sal - f.avg_sal) / m.avg_sal * 100, 2) AS gap_pct,
       f.n AS female_count, m.n AS male_count
FROM role_gender m
JOIN role_gender f ON m.role_id = f.role_id
                  AND m.gender = 'Male' AND f.gender = 'Female'
JOIN job_roles r ON r.role_id = m.role_id
ORDER BY r.job_level;
```

Result

role_name	male_avg	female_avg	gap_pct	female_count	male_count
Intern	310,638	293,916	5.38	172	245
Junior Associate	582,366	542,672	6.82	868	1405
Associate	876,397	828,870	5.42	1004	1588
Senior Associate	1,270,012	1,196,545	5.78	793	1298
Team Lead	1,803,925	1,729,133	4.15	475	719
Manager	2,563,897	2,335,975	8.89	310	456
Senior Manager	3,868,785	3,647,048	5.73	205	336
Director	5,934,282	5,467,230	7.87	116	189

Insight: The gap exists inside every job role (4.2% to 8.9%), so it is NOT explained by women holding lower-level roles. It is widest at Manager level (8.9%) and Director level (7.9%).

Q5. Pay gap by department and city (4-table JOIN)

Business question: Which department and city combinations show the largest gender gap? [4-table JOIN + HAVING]

```
SELECT d.dept_name, c.city_name,
       ROUND(AVG(CASE WHEN e.gender='Male'   THEN e.annual_salary END)) AS male_avg,
       ROUND(AVG(CASE WHEN e.gender='Female' THEN e.annual_salary END)) AS female_avg,
       ROUND((AVG(CASE WHEN e.gender='Male' THEN e.annual_salary END) -
              AVG(CASE WHEN e.gender='Female' THEN e.annual_salary END)) /
              AVG(CASE WHEN e.gender='Male' THEN e.annual_salary END) * 100, 1) AS gap_pct
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
JOIN job_roles  r ON e.role_id = r.role_id
JOIN cities     c ON e.city_id = c.city_id
WHERE r.role_name IN ('Associate', 'Senior Associate')
GROUP BY d.dept_name, c.city_name
HAVING COUNT(CASE WHEN e.gender='Female' THEN 1 END) >= 10
   AND COUNT(CASE WHEN e.gender='Male'   THEN 1 END) >= 10
ORDER BY gap_pct DESC
LIMIT 8;
```

Result

dept_name	city_name	male_avg	female_avg	gap_pct
Customer Support	Jaipur	615,432	494,236	19.7
Data Science	Bengaluru	1,757,629	1,415,331	19.5
Operations	Kochi	855,653	698,888	18.3
Sales	Pune	1,065,446	880,759	17.3
Customer Support	Chandigarh	644,418	536,836	16.7
Operations	Kolkata	823,960	693,673	15.8
Sales	Kochi	876,322	756,579	13.7
HR	Kolkata	869,960	754,192	13.3

Insight: Gaps are far larger in specific pockets: Customer Support in Jaipur (19.7%), Data Science in Bengaluru (19.5%) and Operations in Kochi (18.3%), measured within Associate and Senior Associate roles. These are priority areas for review.

Q6. Employees paid far above their role average

Business question: Who is paid more than 1.5x the average of their own role (possible band breach)? [Subquery in JOIN]

```
SELECT r.role_name, COUNT(*) AS above_1_5x_avg
FROM employees e
JOIN job_roles r ON e.role_id = r.role_id
JOIN (SELECT role_id, AVG(annual_salary) AS role_avg
      FROM employees GROUP BY role_id) ra ON e.role_id = ra.role_id
WHERE e.annual_salary > 1.5 * ra.role_avg
GROUP BY r.role_name
ORDER BY above_1_5x_avg DESC;
```

Result

role_name	above_1_5x_avg
Associate	95
Senior Associate	76
Junior Associate	76
Team Lead	46
Manager	31
Senior Manager	20
Director	13
Intern	11

Insight: 368 employees are paid more than 1.5x their role average. Associate (95), Senior Associate (76) and Junior Associate (76) have the most. These cases need a justification check.

Q7. Top 3 earners in every department

Business question: Who are the three highest-paid employees in each department? [Window: ROW_NUMBER]

```
SELECT dept_name, emp_name, role_name, annual_salary, rank_in_dept
FROM (
    SELECT d.dept_name, e.emp_name, r.role_name, e.annual_salary,
           ROW_NUMBER() OVER (PARTITION BY d.dept_name
                              ORDER BY e.annual_salary DESC) AS rank_in_dept
    FROM employees e
    JOIN departments d ON e.dept_id = d.dept_id
    JOIN job_roles  r ON e.role_id = r.role_id
) t
WHERE rank_in_dept <= 3
ORDER BY dept_name, rank_in_dept;
```

Result (first 3 departments)

dept_name	emp_name	role_name	annual_salary	rank_in_dept
Customer Support	Ramesh Kumar	Director	4,339,000	1
Customer Support	Arjun Iyengar	Director	4,171,600	2
Customer Support	Mohan Yadav	Director	4,170,800	3
Data Science	Naveen Mishra	Director	11,373,800	1
Data Science	Deepak Kapoor	Director	10,542,000	2
Data Science	Faizal Menon	Director	10,537,900	3
Finance	Deepak Naidu	Director	8,109,000	1
Finance	Aarav Rao	Director	8,087,200	2
Finance	Kiran Kumar	Director	7,534,400	3

Insight: Directors are the top three earners in every department. The highest salary is a Data Science Director at Rs 1.14 crore. The first three departments are shown; the full output has 27 rows.

Q8. City pay ranking versus company average

Business question: Which cities pay above or below the company average? [Window: RANK + AVG OVER]

```
SELECT c.city_name,
       ROUND(AVG(e.annual_salary)) AS city_avg,
       ROUND(AVG(AVG(e.annual_salary)) OVER ()) AS overall_avg_of_cities,
       RANK() OVER (ORDER BY AVG(e.annual_salary) DESC) AS pay_rank
FROM employees e
JOIN cities c ON e.city_id = c.city_id
WHERE c.city_name <> 'Unknown'
GROUP BY c.city_name
ORDER BY pay_rank;
```

Result

city_name	city_avg	overall_avg_of_cities	pay_rank
Bengaluru	1,620,063	1,378,128	1
Mumbai	1,614,165	1,378,128	2
Delhi	1,519,678	1,378,128	3
Hyderabad	1,484,021	1,378,128	4
Pune	1,407,632	1,378,128	5
Chennai	1,405,925	1,378,128	6
Kolkata	1,332,842	1,378,128	7
Ahmedabad	1,299,637	1,378,128	8
Kochi	1,276,056	1,378,128	9
Chandigarh	1,255,830	1,378,128	10
Mysuru	1,168,714	1,378,128	11
Jaipur	1,152,970	1,378,128	12

Insight: Bengaluru (Rs 16.2 lakh) and Mumbai (Rs 16.1 lakh) pay most; Jaipur (Rs 11.5 lakh) pays about 29% less than Bengaluru. Six of 12 cities are above the all-city average.

Q9. Share of women in each salary quartile

Business question: Are women under-represented in the top-paid 25%? [CTE + Window: NTILE]

```
WITH q AS (
    SELECT gender, NTILE(4) OVER (ORDER BY annual_salary) AS salary_quartile
    FROM employees
    WHERE gender IN ('Male', 'Female')
)
SELECT salary_quartile,
       COUNT(*) AS employees,
       SUM(CASE WHEN gender = 'Female' THEN 1 ELSE 0 END) AS women,
       ROUND(SUM(CASE WHEN gender = 'Female' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1) AS women_pct
FROM q
GROUP BY salary_quartile
ORDER BY salary_quartile;
```

Result

salary_quartile	employees	women	women_pct
1	2545	1090	42.8
2	2545	949	37.3
3	2545	952	37.4
4	2544	952	37.4

Insight: Women make up 42.8% of the lowest-paid quartile but only 37.4% of the top quartile, which shows they are concentrated at the lower end of the pay scale.

Q10. Does performance close the gap? (rating x gender)

Business question: Do women with the same performance rating earn the same as men? [JOIN + GROUP BY + CASE]

```
SELECT e.performance_rating,
       ROUND(AVG(CASE WHEN gender='Male'   THEN annual_salary END)) AS male_avg,
       ROUND(AVG(CASE WHEN gender='Female' THEN annual_salary END)) AS female_avg,
       ROUND((AVG(CASE WHEN gender='Male' THEN annual_salary END) -
              AVG(CASE WHEN gender='Female' THEN annual_salary END)) /
              AVG(CASE WHEN gender='Male' THEN annual_salary END) * 100, 1) AS gap_pct
FROM employees e
WHERE e.gender IN ('Male','Female')
GROUP BY e.performance_rating
ORDER BY e.performance_rating;
```

Result

performance_rating	male_avg	female_avg	gap_pct
1	1,375,629	1,241,785	9.7
2	1,420,325	1,401,536	1.3
3	1,429,155	1,315,611	7.9
4	1,377,718	1,323,743	3.9
5	1,452,092	1,335,697	8

Insight: Women earn less than men at every performance rating (1.3% to 9.7%). Even top performers (rating 5) show an 8.0% gap, so performance does not explain the difference.

Summary of SQL findings

• **Pay gap is real and consistent:** women earn 6.2% less overall and 4% to 9% less in every role and at every performance rating.

• **Not a role-mix effect:** the gap remains when comparing the same job role, so it points to pay-setting practice rather than seniority.

• **Hot spots:** some department-city pockets show gaps near 20%, led by Customer Support in Jaipur and Data Science in Bengaluru.

• **Weak salary bands:** bands span up to 4.6x and overlap across levels; 368 employees earn above 1.5x their role average.

• **Geography and department matter:** Data Science pays 2.2x Customer Support, and Bengaluru pays about 29% more than Jaipur.

• **Women are concentrated in the lowest quartile** (42.8%) and under-represented in the top (37.4%).

• **Next step:** Step 4, Python EDA, to visualise these patterns with distributions, correlations and segmentation.